Advanced Data Analysis

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Contents

Chapter 1

Prerequisites

```r
install.packages("bookdown")
# or the development version
# devtools::install_github("rstudio/bookdown")
```

This book's skeleton is based on several courses. Hence, I'd like to credit professors accordingly.

Course | Professor :-:|:-" Analyzing Unstructured Data | Kunpeng Zhang Deep Learning | Yann LeCun

Chapter 2

Introduction

3 types of data

- Structured Data (tables form)
- Semi-structured Data (e.g., XML, JSON, HTML)
- Unstructured Data
    - Texts
    - Images: Videos, photos,
    - Networks: Social Networks

Chapter 3

Machine Learning

This section based on notes in "Analyzing Unstructured Data" course taught by professor Kunpeng Zhang.

- ML is concerned with the the development, the analysis, and the application of algorithm that allow computers to learn

- Learning:
  - A computer learns if it improves its performance at some tasks with experience (i.e., data)

  - Extracting a model of system from the sole observation (or the simulation)

  - A model = some relationships between the variables used to describe the system.

- Goals:
  - Make prediction

  - Understand the underlying mechanisms.

- Learning if useful when:
  - Human expertise does not exist (navigating on Mars)

  - Humans are unable to explain their expertise (speech recognition)

  - Solution changes in time (routing on a computer network)
    Solution needs to be adapted to particular cases (user biometrics)

- Applications in different domains:

  - Retail: Market basket analysis, customer relationship management (CRM)
  - Finance: Credit scoring, fraud detection.
  - Manufacturing: Optimization, troubleshooting
  - Medicine: medical diagnosis
  - Telecommunications: Quality of service optimization, routing
  - Bioinformatics: Motifs, alignment WEb mining, search engines.

- Steps
  - Data generation: data types, sample size
  - Preprocessing: normalization, missing values, feature selection/extraction
  - Machine Learning: Hypothesis, choice of learning paradigm/algorithm.
  - Hypothesis validation: cross-validation, model deployment.

- Types of machine learning:
  - Supervised: manually label variables
  - Unsupervised:

- Training and testing
  - Training is the process of making the machine learn
  - No free lunch:
    * Training set and testing set come from the same distribution
    * Need to make some assumptions.

- Factors affecting the performance:
  - Types of training provided
  - The form and extend of prior knowledge
  - Type of feedback provided
  - The learning algorithms used

- Two factors:

  - Modeling
  - optimization

- Algorithms:
  - ML depends on algorithms.
  - The algorithms control the search to find and build the knowledge structures
  - The learning algorithms should extract useful information from training examples.

- Types of learning (algorithms):
  - Supervised learning ($\{X_n \in R^d, y_n \in R\}_{n=1}^N$)
    * Prediction
    * Classification (discrete labels), regression (real values)
  - Unsupervised learning ($\{X_n \in R^d\}_{n=1}^N$)
    * Clustering
    * Probability distribution estimation
    * Finding association (in features)
    * Dimension reduction
  - Semi-supervised learning
  - Reinforcement learning
    * Decision making (robot, chess machine)

3.1 Supervised Learning

From the data, find a function f of the inputs that best approximates the outputs.

$$\hat{Y} = f(X_1, .., X_n)$$

- Predictive: Make predictions for a new object described by its attributes
- Informative: help understand the relationship between inputs and outputs

Divide data into

- Training set: train model
- validation set: tune hyper parameters
- Test set: evaluation of the model for generalization error.

Learning algorithm

- defined by
  - a family of candidate models (= **hypothesis space H**)
  - a quality measure for a model
  - an optimization strategy
- takes as input a leaning sample and outputs a function h in H of maximum quality

Model selection

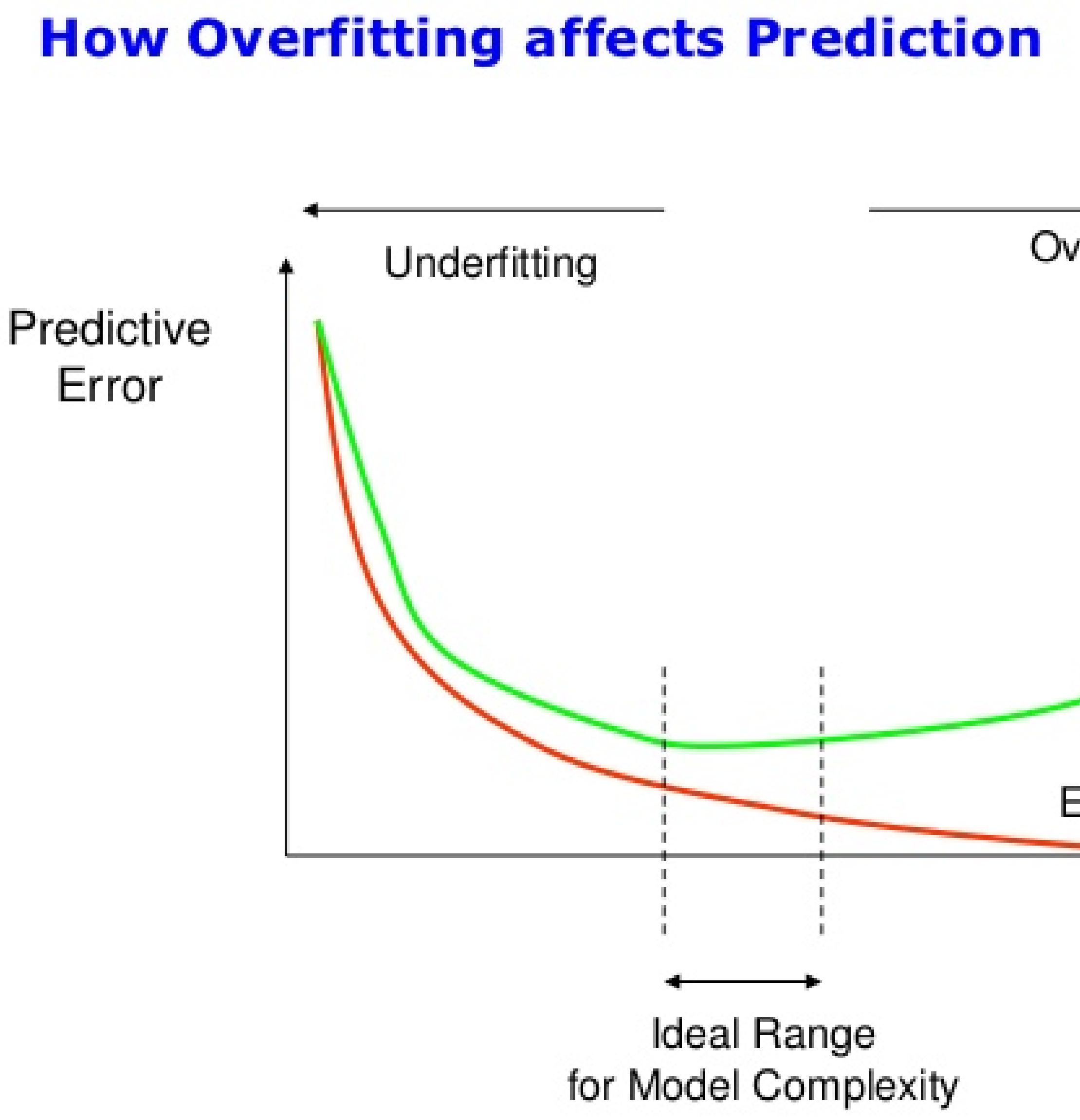
- Why not choose the model that minimizes the error rate on the learning sample? (**re-substitution error**)
- How well are you going to predict future data drawn from the same distribution? (**generalization error**)
- Evaluation on test set
  1. Randomly choose a percentage (e.g., 30%) of the data to be in a test set
  2. The remainder is training set
  3. Learn the model from the training set
  4. Estimate its future performance on the test set.
- Problems:
  - overfits: occurs when the learning algorithm starts fitting noise.
  - unfits
- Evaluation on test set:
  - Good:
    * Very simple
    * Computationally efficient
  - Bad:

    * Waste data
    * very unstable when database is small

- Another evaluation: **Leave-one-out cross validation**
  - For k = 1 to N
    * remove the k-th object from the learning sample
    * learn the model on the remaining objects
    * apply the model to get a prediction for the k-th object
  - report the proportion of mis-classified objects
- Good:
  - Do not waste the data (you get an estimate of the best method to apply to N-1 data)
- Bad:
  - Computationally intensive (train N models)
  - high variance.
- Another evaluation: **K-fold cross validation**
  - randomly partition the dataset into k subsets (e.g., 10)
  - For each subset:
    * learn the model on the objects that are not in the subset
    * compute the error rate on the points in the subset. Report the mean error rate over the k subsets.
  - When k = the number of objects -> leave-one-out-cross validation.

Rule-of-thumb for evaluation

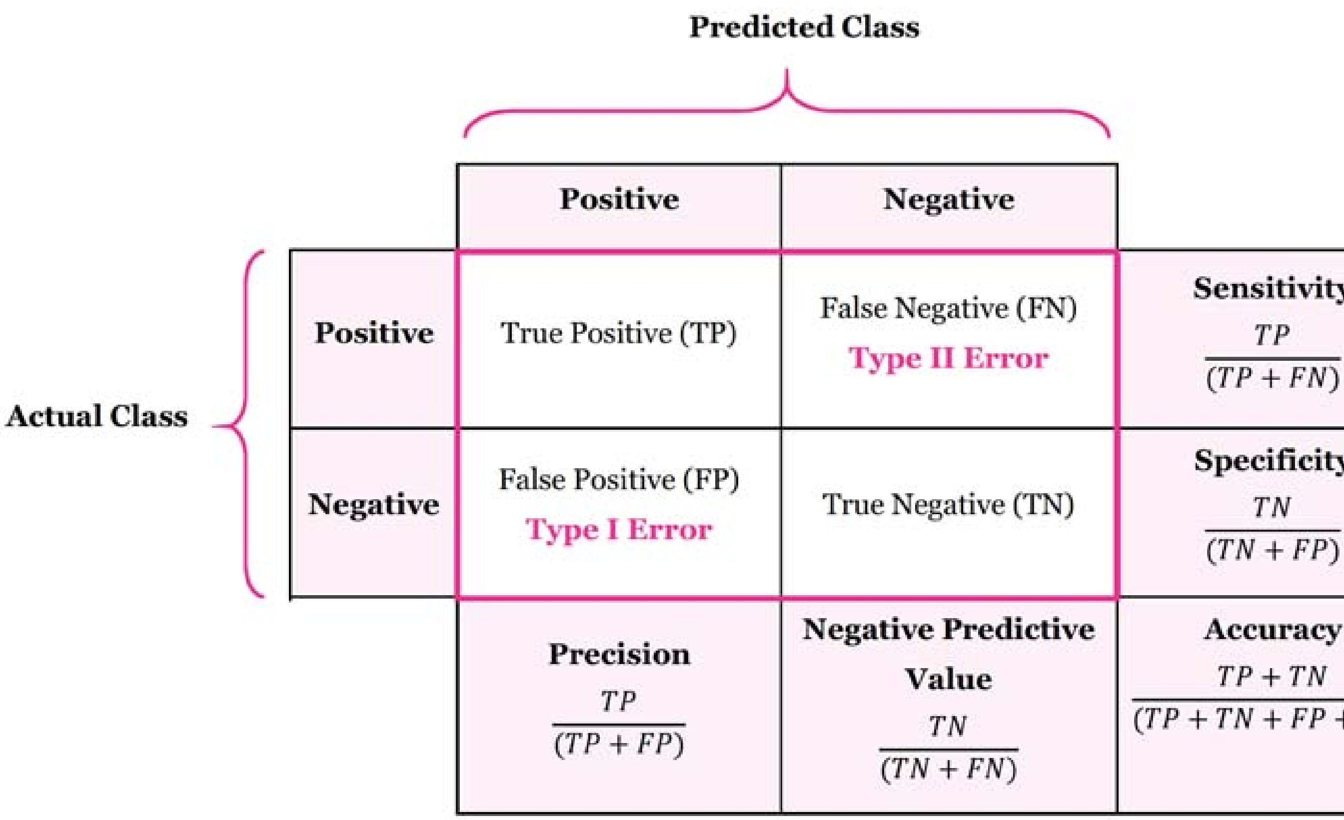
- Lots of data (>1000) test set validation
- small data (100-1000): 10-fold CV
- very small data (<100) leave one out CV.


How Overfitting affects Prediction
Underfitting
Predictive
Error
Ideal Range
for Model Complexity

Use the testing set to estimate the performance of the final model

Performance measures

- The error rate is no the only way to assess a predictive model.
- In binary classification, results can be summarized in a contingency table (i.e., confusion matrix)



Various criteria

- Error

ROC and precision/recall curves

- Each point corresponds to a particular choice of the decision threshold.

Also known as robustness check

You want large area under the curve

Compassion of learning models

- Three main criteria:
  - Accuracy: Measures by the generalization error (estimated by CV)
  - Efficiency: Computing times and scalabiility for learning and testing
  - Interpretability: Comprehension brought by the model about the input-output relationship.
- Depends on the application, you have to make trade-off.

Learning Techniques

- Supervised learning categories and techniques
  - Linear classifier (numerical functions):
    * Perceptron
    * Logistic regression
    * Support vector machine (SVM): linear to nonlinear: feature transofrm and kernel function
    * Ada-line
    * Multi-layer perceptron (MLP)
  - Parametric (Probabilistic functions)
    * Naive Bayes, Gaussian discriminant analysis (GDA), Hidden Markov models (HMM), probabilistic graphical models
  - Non-parametric (Instance-based functions)
    * K-nearest neighbors, Kernel regression, Kernel density estimation, local regression.
  - Non-metric (symbolic functions)
    * Classification and regression tree (CART), decision tree
  - Aggregation
    * Bagging (bootstrap + aggregation), Adaboost, Random forest.
- Unsupervised learning categories and techniques:
  - Clustering
    * K-means clustering
    * Spectral clustering
  - Density Estimation
    * Gaussian mixture model (GMM)

    * Graphical models
  - Dimensionality reduction
    * Principal component analysis (PCA)
    * Factor analysis
- Semi-supervised learning
  - goal: exploit both labeled and unlabeled data to build better models than using each one alone
  - Self-training:
    * Iteratively label some unlabeled examples with a model learned from the previously labeled examples
    * Active learning.
- Reinforcement learning
  - Learning from interactions
  - Goal: learn to choose sequence of actions (=policy) that maximizes $r_0 + \gamma r_1 + \gamma^2 r_2 + ...$ where $0 \leq \gamma < 1$

3.2 Unsupervised Learning

3.2.1 Clustering

Clustering is used to find similar groups in data (i.e., clusters)

No class value denoting an a priori grouping (hence, Unsupervised Learning)

Clustering algorithm

- Partitional clustering
- Hierarchical clustering

A distance (similarity, or dissimilarity) function

Clustering quality

- Inter-clusters distance -> maximized
- Intra-clusters distance -> minimized

The quality of a clustering result depends on

- The algorithm

- the distance function
- the application

3.2.1.1 Similarity Measures

It's your choice of how to define similarity

(Dis)similarity measure

- Equivalently, we can use dissimilarity measures
- Jagota defines a dissimilarity measure as a function f(x,y) such that f(x,y) > f(w,z) if and only if x is less similar to y than w is to z.
  Also known as **pair-wise** measure

Distance functions

Different distance functions for

- Different types of data
  - Numeric data
  - Nominal data
- Different applications

Numeric data

- $dist(x_i, x_j)$,where
  - $dist(x_j, x_j) = ((x_{i1} - x_{j1})^h + (x_{i2} - x_{j2})^h + ... + (x_{ir} - x_{jr})^h)^{\frac{1}{h}}$
  - x_i and x_j are data points (vectors)
- Euclidean distance
  - If h = 2, it is the Euclidean distance: $dist(x_j, x_j) = ((x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + ... + (x_{ir} - x_{jr})^2)^{\frac{1}{2}}$
  - Weighted Euclidean distance: $dist(x_j, x_j) = (w_1(x_{i1} - x_{j1})^2 + w_2(x_{i2} - x_{j2})^2 + ... + w_r(x_{ir} - x_{jr})^2)^{\frac{1}{2}}$
  - Squared Eulidean distance: to place progressively greater weight on data points that are further apart: $dist(x_j, x_j) = (x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + ... + (x_{ir} - x_{jr})^2$
- Manhattan distance
  - If h = 1, it is the Manhattan distance: $dist(x_j, x_j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + ... + |x_{ir} - x_{jr}|$
- Minkowski distance (h is positive integer)
  - $dist(x_j, x_j) = ((x_{i1} - x_{j1})^h + (x_{i2} - x_{j2})^h + ... + (x_{ir} - x_{jr})^h)^{\frac{1}{h}}$
  - 1st dimension, 2nd dimension, etc.

- Chebychev distance: define 2 data points as "different" if they are different on any one of the attributes
  - $dist(x_j, x_j) = max(|x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + ... + |x_{ir} - x_{jr}|)$

Binary and nominal Attributes

- Binary attribute: has 2 values or states but no ordering relationship (e.g., gender)
- We use a confusion matrix to introduce the distance function/measures
- Let the i-th and j-th data points be X_i and x_j (vectors)

slide 79

3.2.1.2 K-mean Algorithm

K-means is a partitional clustering algorithm.

Let the set of data points (or instances) D to be $(x_1, ..., x_n)$, where $x_i = (c_{i1}, ..., x_{ir})$ is a vector in a real-valued space $X \subseteq R^r$ and r is the number of attributes (dimensions) in the data

The k-means algorithm partitions the given data into k clusters

- Each cluster has a cluster center (centroid)
- k is specified by user.

Steps

1. Randomly choose k data points (seeds) to be the initial centroids (cluster centers)
2. Assign each data point to the closest centroid
3. Re-compute the centroids using the current cluster memberships
4. If a convergence criterion is not met, go back to 2.

Convergence Criterion

1. No(or minimum) re-assignments of data points to different clusters,
2. No (or minimum) change of centroids, or
3. minimum decreases in the sum of squared error (SSE)

$$SSE = \sum_{j=1}^{k} \sum_{x \in C_j} dist(x.m_j)^2$$

- C is the j-th cluster,m is the centroid of cluster C_j (the mean vector of all the data points in C). and dist(x,m) is the distance between data point x and centroid m_j.

Pros

- Simple: easy to understand and implement

- Efficient: time complexity: O(tkn), where n is the number of data points, k is the number of clusters, and t is the number of iterations.

- Since both ka nd t are small, k-means is considered a linear algorithm

- Note: it terminates at a local optimum if SSE is used. The global optimum is hard to find due to complexity.

Cons

- The algorithm is only applicable if the mean is defined.
  - For categorical data, k-mode

- The user needs to specify k

- The algorithm is sensitive to outliers.
  - outliers could be errors in the data recording or some special data points with very different values.

Solution

Chapter 4

Methods

We describe our methods in this chapter.

Chapter 5

Applications

Some *significant* applications are demonstrated in this chapter.

5.1 Example one

5.2 Example two

Chapter 6

Final Words

We have finished a nice book.

Part I

ADVANCED

Chapter 7

Deep Learning

7.1 Overview

- What is deep learning? Input to output via layers of representation
- What are layers? A layer is a geometric transformation function on the data that goes through it Weights determine the data transformation behavior of a layer

Learning Representation

- Transforming input data into useful representation

The “deep” in deep learning

- multiple layers
- Other possibly more appropriate names for the field:
  - Layered representations learning
  - Hierarchical representations learning
  - Chained geometric transformation learning

New problem domains for R:

- Computer vision
- Computer speech recognition
- Reinforcement learning applications

How do we train deep learning models?

- Basic of machine learning algorithms

- Machine learning vs. statistical modeling
- MNIST example
  - Model definition in R
  - Layers of representation
- The training loop

Machine learning algorithms
Learning model parameters via exposure to many example data points

Statistics: Often focused on inferring the process by which data is generated
Machine Learning: Principally focused on predicting future data

Deep learning frontiers

- Computer vision
- Natural language processing
- Time series
- Biomedical

7.2 Tensor Flow and R

TensorFlow APIs

- Keras API
- Estimator API
- Core API

7.2.1 R packages

TensorFlow APIs * keras: Interface for neural networks, with a focus on enabling fast experimentation. * tfestimators: Implementations of common model types such as regressors and classifiers.
* tensorflow: Low *level interface to the TensorFlow computational graph.* tf-datasets: Scalable input pipelines for TensorFlow models.

Supporting Tools * tfruns : Track , visualize, and manage TensorFlow training runs and experiments * tfdeploy : Tools designed to make exporting and serving TensorFlow models straightforward. * cloudml : R interface to Google Cloud Machine Learning Engine.

R interface to Keras

- Step by step example
- Keras layers
- Compiling models
- Losses, Optimizers, and Metrics
- More examples

7.2.2 Keras layers

65 layers available

- Dense layers: classic "fully connected" neural network layers
- Convolutional layers: "Filters for learning local patterns in data
- Recurrent layers: Layers that maintain state based on on previously seen data
- Embedding layers: Vectorization of text that reflects semantic relationships between words

7.3 Compiling models

Model compilation prepares the model for training by:
1. Converting the layers into a TensorFlow graph
2. Applying the specified loss function and optimizer
3. Arranging for the collection of metrics during training

7.4 Losses, Optimizers, and Metrics

All available at Keras for R cheatsheet

Example of Image classificaiton

7.5 NYU

Materials in this chapter is based on NYU Deep Learning

Deep in Deep Learning means multi-layers. "A deep network has several layers and uses them to build a hierarchy of features of increasing complexity"

Supervised Learning (= Function Optimization): With inputs and outputs, you train the machine to tweak its parameter to have the correct outputs from inputs. In "traditional machine learning", you have to engineer **feature extractor**, but in deep learning you can multi-layers feature extractor can be trained. And all layers are non-linear, because if they are linear, then all layers would be collapse into one layer.

Natural Data is compositional. Hence, it is efficiently representable hierarchically.

Revolutions in Deep Learning:

- Speech Recognition: 2010
- Image Recognition: 2013
- Natural Language Processing (NLP): 2015

Deep Learning = Learning Representations/ Features Representation means you turn raw data into something useful

Image Recognition:
Pixel, edge, texton, motif, part, object

Text
Character, group, word group, clause, sentence, story

Speech
Sample, spectral band, sound, phone, phoneme, word.

Difference between Deep Learning Support-Vector Machines (SVM)

SVM is a 2-layer neural net, where

- layer 1 = templates
- layer 2 = linear classifier.

where Deep Learning is "A deep network has several layers and uses them to build a hierarchy of features of increasing complexity"

Deep Machines are more efficient than SVM in representing certain classes of functions.

The Manifold Hypothesis "states that real-world high-dimensional data lie on low-dimensional manifolds embedded within the high-dimensional space"

Difference between Deep Learning and PCA (Principal Components Analysis) is that Deep learning uses non-linear function, while PCA uses linear function.

Chapter 8

Interpretable Machine Learning

https://christophm.github.io/interpretable-ml-book/ https://en.wikipedia.org/wiki/Explainable_artificial_intelligence https://arxiv.org/pdf/1905.08883.pdf https://arxiv.org/abs/2004.14545 https://www.kaggle.com/learn/machine-learning-explainability https://towardsdatascience.com/interperable-vs-explainable-machine-learning-1fa525e12f48

8.1 SHAP

https://github.com/slundberg/shap https://meichenlu.com/2018-11-10-SHAP-explainable-machine-learning/ https://bjlkeng.github.io/posts/model-explanability-with-shapley-additive-explanations-shap/

8.2 Video Classification

https://hackernoon.com/continuous-video-classification-with-tensorflow-inception-and-recurrent-nets-250ba9ff6b85 https://cloud.google.com/video-intelligence https://arxiv.org/pdf/1609.06782.pdf https://www.tubefilter.com/2019/02/12/taxonomy-youtube-videos-develop-original-content-that-works/ https://medium.com/swlh/converting-news-into-video-stories-5d8d4fc5a32b https://towardsdatascience.com/introduction-to-video-classification-6c6acbc57356 https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0228579

8.3 Image Memorability

(AMNET)

https://www.researchgate.net/publication/335603628_Image_Memorability_Using_Diverse_Visual_Features_and_Soft_Attention https://www.researchgate.net/publication/308812354_Deep_Learning_for_Image_Memorability_Prediction_the_Emotional_Bias https://arxiv.org/pdf/2005.02969.pdf https://github.com/dazcona/memorability https://arxiv.org/abs/1804.03115 https://www.researchgate.net/publication/324387176_AMNet_Memorability_Estimation_with_Attention http://ceur-ws.org/Vol-2283/MediaEval_18_paper_24.pdf https://jov.arvojournals.org/article.aspx?articleid=2771173 https://openaccess.thecvf.com/content_cvpr_2018/papers/Fajtl_AMNet_Memorability_Estimation_CVPR_2018_paper.pdf https://ieeexplore.ieee.org/document/8578764

https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=7382237&tag=1 https://people.umass.edu/~yangzhou/

http://people.csail.mit.edu/khosla/projects/regionmem/ https://github.com/adikhosla/feature-extraction https://expertprogrammanagement.com/2019/12/elaboration-likelihood-model/

Chapter 9

Research Methods

9.1 But-for World

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